

BIRR, Ireland

WMO: 039650

Lat: 53.083N

Lon: 7.883W

Elev: 72

StdP: 100.46

Time Zone: 1.00 (IRL)

Period: 98-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.9	-1.5	-4.3	2.6	-0.4	-3.0	3.0	0.4	11.4	7.8	10.5	8.4	1.5	190	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	23.0	16.8	21.3	16.2	20.0	15.7	18.2	20.8	17.3	19.8	16.5	18.9	3.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.3	19.4	16.3	11.7	18.5	15.5	11.1	17.7	51.5	21.2	49.0	19.9	46.6	19.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.6	7.7	DB	-6.2	26.3	1.1	1.8	-7.0	27.5	-7.7	28.6	-8.3	29.6	-9.1	30.8	(4)
(5)			WB	-6.3	20.0	1.2	1.1	-7.2	20.8	-7.9	21.5	-8.5	22.1	-9.4	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	5.8	5.7	6.9	8.7	11.4	13.9	15.4	15.3	13.5	10.3	7.7	5.8
	DBStd	4.43	2.94	2.68	2.65	2.55	2.29	2.36	2.08	1.90	2.32	2.75	2.69	3.06
	HDD10.0	662	132	123	99	54	11	1	0	0	2	30	77	132
	HDD18.3	3031	389	355	353	289	216	135	96	96	145	249	319	388
	CDD10.0	680	1	1	4	14	54	117	167	165	108	39	8	2
	CDD18.3	8	0	0	0	0	0	1	4	3	1	0	0	0
	CDH23.3	45	0	0	0	0	0	6	26	12	1	0	0	0
	CDH26.7	4	0	0	0	0	0	0	3	1	0	0	0	0
Wind	WSAvg	3.9	4.5	4.2	4.2	4.0	3.8	3.5	3.3	3.3	3.4	3.9	3.9	4.2
Precipitation	PrecAvg	825	79	57	61	57	59	61	61	79	73	82	77	82
	PrecMax	1113	150	169	100	120	121	160	145	178	147	145	152	166
	PrecMin	617	9	2	13	12	4	7	12	6	2	32	26	24
	PrecStd	103	35	34	25	26	30	32	29	37	36	32	27	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.5	15.6	18.8	22.3	24.2	26.3	24.9	22.1	17.7	14.6	13.7
		MCWB	11.5	11.5	11.3	13.1	15.3	16.3	19.0	19.2	17.7	14.9	12.8	12.2
	2%	DB	11.8	11.7	13.5	16.3	19.8	22.0	23.3	21.7	20.2	16.4	13.5	12.2
		MCWB	10.5	10.1	10.3	11.6	14.2	16.0	17.3	16.9	16.5	14.1	12.1	11.1
	5%	DB	11.1	10.9	12.3	14.7	17.7	20.3	21.0	20.3	18.6	15.4	12.7	11.3
		MCWB	9.9	9.3	9.9	10.8	12.9	15.3	16.4	16.3	15.5	13.5	11.5	10.2
	10%	DB	10.3	10.0	11.4	13.2	16.0	18.6	19.3	19.0	17.6	14.5	11.8	10.4
		MCWB	9.2	8.5	9.3	10.0	12.1	14.4	15.5	15.4	14.9	12.8	10.7	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.7	12.2	13.7	16.1	18.1	19.6	19.9	18.6	15.5	13.4	12.4
		MCDB	12.6	12.5	14.0	17.3	20.5	21.4	24.4	24.3	20.8	16.8	14.0	13.2
	2%	WB	10.8	10.6	11.4	12.5	14.9	16.7	18.1	18.2	17.2	14.7	12.5	11.3
		MCDB	11.5	11.5	12.9	15.2	18.7	20.3	21.7	20.1	19.0	15.9	13.2	12.1
	5%	WB	10.1	9.6	10.5	11.4	13.7	15.8	17.1	17.3	16.3	13.9	11.6	10.5
		MCDB	10.9	10.6	11.8	13.8	16.5	19.3	20.0	19.1	18.1	15.1	12.5	11.2
	10%	WB	9.3	8.7	9.5	10.4	12.7	14.9	16.3	16.3	15.3	13.0	10.9	9.4
		MCDB	10.2	9.9	11.0	12.7	15.1	17.9	18.7	18.1	17.1	14.3	11.7	10.3
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.2	7.0	7.9	8.2	7.9	7.4	7.4	7.4	6.7	5.7	5.3
		MCDBR	5.6	6.2	7.7	9.9	10.9	10.6	10.2	9.1	8.3	6.7	5.4	5.7
		MCWBR	5.3	5.3	5.3	5.9	6.2	5.8	5.2	5.0	5.0	4.9	4.9	5.4
	5% WB	MCDBR	5.7	6.0	6.3	8.2	9.7	9.5	8.7	7.4	7.1	6.3	5.2	5.7
		MCWBR	5.5	5.6	5.1	5.4	5.9	5.7	5.1	4.9	5.3	5.3	5.0	5.6
Clear-Sky Solar Irradiance	taub	0.337	0.346	0.369	0.377	0.388	0.375	0.379	0.360	0.355	0.343	0.330	0.327	
	taud	2.245	2.301	2.270	2.279	2.285	2.352	2.348	2.411	2.424	2.410	2.303	2.210	
	Ebn at Noon	615	746	809	853	861	876	864	863	821	746	623	543	
	Edh at Noon	66	84	105	118	123	117	115	102	90	74	61	57	
All-Sky Solar Radiation	RadAvg	0.71	1.40	2.50	3.86	4.84	5.06	4.58	3.87	2.87	1.68	0.88	0.54	
	RadStd	0.07	0.12	0.25	0.37	0.34	0.47	0.39	0.31	0.18	0.14	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page